Retrieval-Augmented Simplification of Sectioned Biomedical Abstracts

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Motivation

The Cochrane Database of Systematic Reviews pairs every technical abstract with a plain-language summary (PLS), intended to be readable without medical training. This can be tedious work, making an automated approach appealing. Generating a PLS from an abstract is, however, non-trivial, as large parts of the abstract have to be rephrased into simpler language, and additional background content that is not in the abstract has to be inserted into the PLS [3].

Contributions

- We propose a RAG pipeline for the generation of Cochrane PLS that uses abstract-PLS section pairs from topically related reviews as in-context examples.
- We split each retrieved PLS section into its rephrased and inserted sentences, and present both parts separately in the prompt, showing the model which source content to keep and what background to add.
- We show that our pipeline outperforms four strong baselines on the Cochrane-sections test set, with the same gain trend across three different LLM families.

The Cochrane-sections dataset

Cochrane-sections [1] divides the PLS of 7.7K Cochrane reviews into the same seven sections as their abstracts. A PLS sentence is rephrased if it has a counterpart in the abstract, and inserted if it does not.

Abstract section (*Background*, CD001823)
Lumbar supports are used in the treatment of low-back pain patients, to prevent the onset of low-back pain (primary prevention) or to prevent recurrences of a low-back pain episode (secondary prevention).

Rephrased PLS sentence
Lumbar supports (also called braces or corsets) are used in the prevention and treatment of low-back pain.

Inserted PLS sentence
This review is important because low-back pain is very common.

Retrieval-augmented simplification pipeline

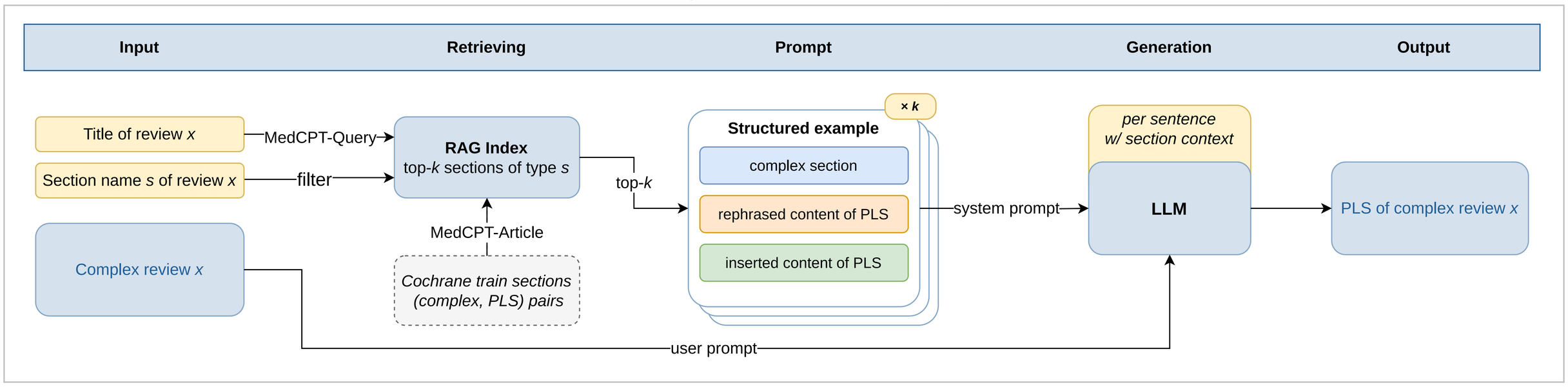


Figure 1. Overview of the RAG pipeline for the simplification of sectioned biomedical abstracts. For each section of a complex abstract that the LLM is asked to simplify, we retrieve the top-k most similar reviews from the RAG index and take from each the matching section. Every retrieved example is then put in the prompt as the complex abstract section together with its PLS section, which is split into rephrased and inserted sentences.

Main results

System	SMART F1 ↑	FKGL ↓	SARI ↑ add	SARI ↑ keep	SARI ↑ all
Input	35.16	13.00	0.00	23.16	7.72
Reference	100.00	11.39	100.00	100.00	100.00
LLM	31.01	10.25	6.46	33.65	43.69
RAG rand-full	31.52	10.30	6.60	34.23	43.78
RAG rand-split	31.58	10.51	6.51	34.33	43.74
RAG retrieved-full	32.53	**10.19**	7.22	35.64	44.26
RAG retrieved-split	**33.21**	10.29	**7.83**	**37.45**	**45.17**†

Table 1. Simplification results on the Cochrane-sections test set ($k = 4$, Qwen2.5-32B-Instruct), averaged over three runs with three different seeds. † significantly better than every other system on per-review SARI (paired bootstrap, $p < 0.01$).

Random abstract-PLS pairs have little effect on simplification quality. Retrieving closely related section examples clearly improves it, and splitting them into rephrased and inserted content boosts it further, driven by the keep and add components.

Per-section SARI

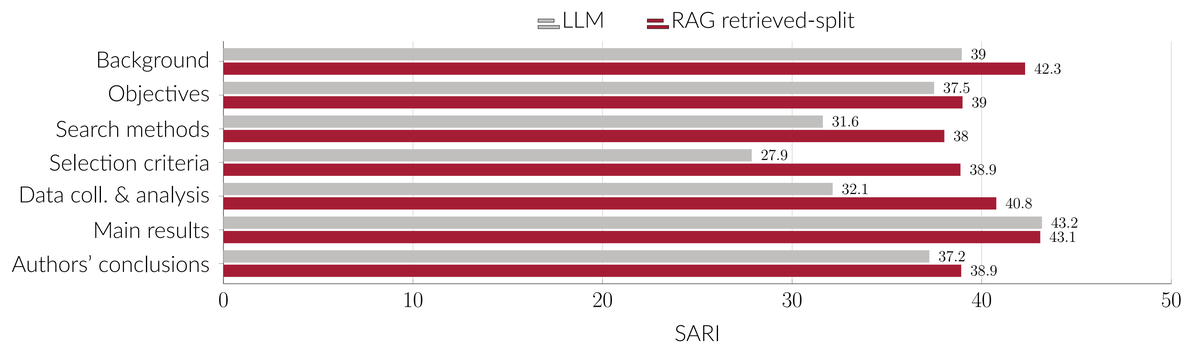


Figure 2. Per-section SARI on the Cochrane-sections test set, averaged over three runs.

The gains are largest for the sections the LLM baseline handles worst without examples. *Main results*, the section the LLM baseline already handles best, stays roughly unchanged. On the CLEF 2026 SimpleText test set [2], the largest gains are again on *Search methods* (+28.4) and *Selection criteria* (+15.3).

Qualitative example (CD009170, *Selection criteria*)

Complex

Randomised controlled trials (RCTs) of blunt versus sharp suture needles for preventing needle stick injuries among surgical staff measured as glove perforations or self-reported needle stick injuries.

LLM (SARI 25.9)

This review looked at studies comparing blunt and sharp suture needles to see which ones help prevent needle stick injuries in surgeons.

RAG retrieved-split (SARI 38.9)

We looked at studies comparing blunt suture needles to sharp ones to see if blunt needles help prevent needle stick injuries among surgical staff. Needle stick injuries were measured by checking for holes in gloves or by asking staff if they had been injured.

Generalization

System	Qwen-32B	Gemma-27B	Llama-70B
LLM	43.69	42.79	43.33
RAG retr.-split	**45.17**	**44.10**	**44.96**
Δ	+1.48	+1.31	+1.63

Table 2. Document-level SARI on the Cochrane-sections test set for three LLM families.

System	Document	Section mean
LLM	46.70	32.89
RAG retr.-split	**47.98**	**39.76**
Δ	+1.28	+6.87

Table 3. SARI on the CLEF 2026 SimpleText test set, full document and section mean.

All three model families gain 1.3 to 1.6 SARI points over their non-RAG LLM baseline. The pipeline also outperforms the non-RAG baseline on the larger, held-out CLEF 2026 SimpleText test set.

Takeaways

- Retrieved section pairs from topically related reviews improve PLS generation, while random pairs do not.
- Splitting each retrieved PLS section into rephrased and inserted sentences boosts quality further.
- The gains hold across three LLM families and on CLEF 2026 [2].

Human evaluation

52.6%
RAG preferred overall
95% CI $[41.6, 63.3]$

83.3%
RAG preferred on *Background*
95% CI $[55.2, 95.3]$

This preference is not statistically significant, and inter-annotator agreement is negligible (Fleiss' $\kappa = 0.08$). Participants who preferred the LLM baseline valued its brevity. Participants who preferred the RAG pipeline valued its completeness, commenting that it "explains terms in the text".

References

[1] Jan Bakker and Jaap Kamps. Section-level simplification of biomedical abstracts. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 13819–13833, Suzhou, China, November 2025. Association for Computational Linguistics.

[2] Liana Ermakova, Hosein Azarbonyad, Jan Bakker, Gautam Kishore Shahi, Benjamin Vendeville, and Jaap Kamps. CLEF 2026 SimpleText track: Simplify scientific text (and nothing more). In Ricardo Campos et al., editors, *Advances in Information Retrieval: 48th European Conference on Information Retrieval, ECIR 2026*, volume 16486 of *Lecture Notes in Computer Science*, pages 215–224, Cham, 2026. Springer.

[3] Yue Guo, Wei Qiu, Gondy Leroy, Sheng Wang, and Trevor Cohen. Retrieval augmentation of large language models for lay language generation. *Journal of Biomedical Informatics*, 149:104580, 2024.